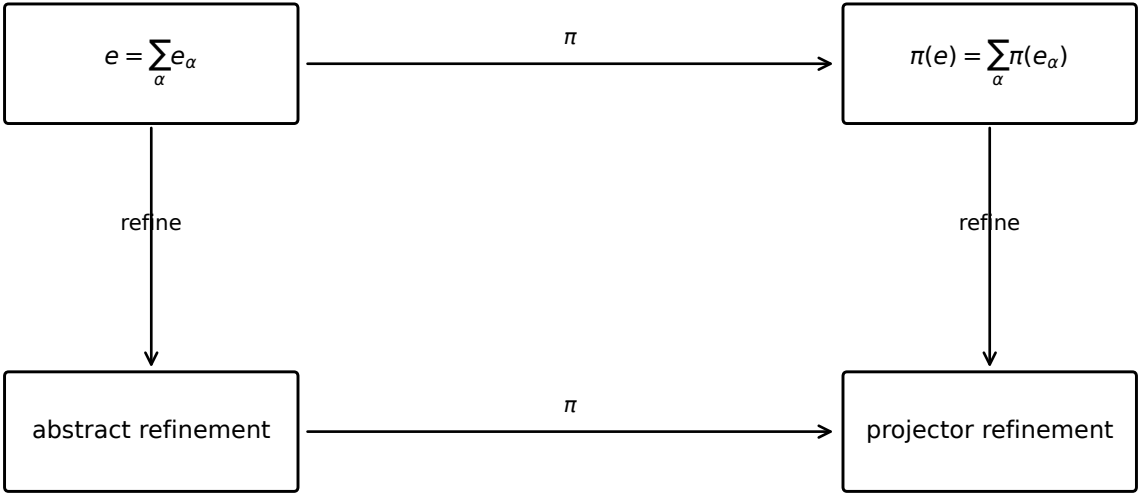representation and refinement commute